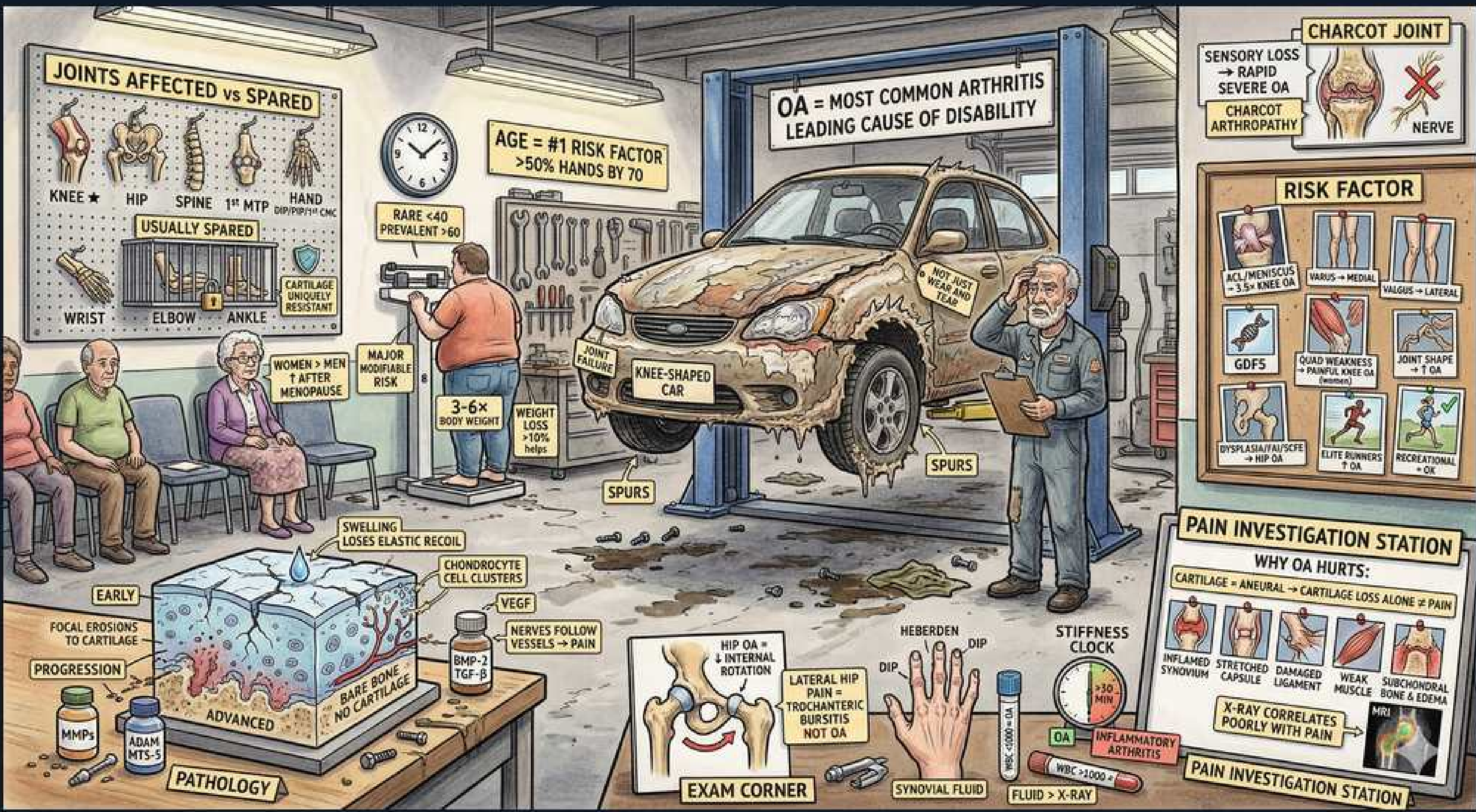
Osteoarthritis — The Repair Garage (Joints, Risk Factors & Pathology)



1. Most common arthritis

WHAT YOU SEE

Banner 'OA = MOST COMMON ARTHRITIS — leading cause of disability'

WHAT IT ENCODES

Osteoarthritis is the most common joint disease worldwide and the leading cause of chronic disability in adults. It is a degenerative, non-inflammatory (or low-grade inflammatory) arthropathy driven by mechanical load plus active cartilage breakdown — fundamentally different from autoimmune RA. Prevalence rises steeply with age: over half of people have radiographic hand OA by age 70. On boards, an older patient with use-related joint pain and brief stiffness defaults to OA unless red flags push you elsewhere.

BOARD TRAP

WRONG answer: 'rheumatoid arthritis is the most common arthritis.' RA prevalence is only ~0.5–1%, whereas symptomatic OA affects a large fraction of older adults. The discriminator is mechanism — OA is degenerative/mechanical with cool, bony joints and brief stiffness; RA is autoimmune synovitis with warm, boggy joints and prolonged stiffness.

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2. Age = #1 risk factor

WHAT YOU SEE

Clock, 'AGE = #1 RISK FACTOR', '>50% hands by 70'

WHAT IT ENCODES

Increasing age is the single strongest risk factor for OA because cumulative mechanical wear, declining chondrocyte repair capacity, and accumulated matrix damage outpace cartilage regeneration. Other key risk factors: female sex (especially post-menopause), obesity (knee), prior joint trauma/surgery, occupational/repetitive loading, and genetic predisposition. Radiographic hand OA exceeds 50% by age 70.

BOARD TRAP

WRONG answer: attributing OA in a 25-year-old to ordinary primary OA. Primary OA is uncommon in the young — early-onset or atypical-joint OA should trigger a search for SECONDARY causes (prior trauma, hemochromatosis, ochronosis, acromegaly, avascular necrosis, or a developmental hip disorder). The discriminator is age plus joint distribution.

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3. Joints affected: DIP, knee, hip, 1st CMC, 1st MTP

WHAT YOU SEE

Board 'JOINTS AFFECTED vs SPARED': knee, hip, spine, 1st CMC, hand

WHAT IT ENCODES

OA preferentially targets weight-bearing and high-use joints: DIP and PIP joints of the hands, the 1st carpometacarpal (thumb base), knees, hips, lumbar/cervical spine (facets), and 1st metatarsophalangeal joint (great toe — hallux rigidus). The thumb-base 1st CMC involvement produces the classic 'squared' hand. Memorizing this favored distribution is the fastest way to separate OA from RA on a vignette.

BOARD TRAP

WRONG answer: choosing RA when a patient has DIP nodules and thumb-base pain. DIP and 1st CMC involvement is HIGHLY characteristic of OA, not RA — RA classically spares the DIPs. The discriminator: DIP + thumb base = OA; MCP + wrist = RA.

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4. Joints SPARED: MCP, wrist, elbow, ankle

WHAT YOU SEE

'USUALLY SPARED — wrist, elbow, ankle, MCP'

WHAT IT ENCODES

OA usually SPARES the MCPs, wrists, elbows, shoulders, and ankles. When these 'non-OA' joints are involved, look for another diagnosis: MCP + wrist OA-like changes suggest HEMOCHROMATOSIS (especially 2nd/3rd MCP hook osteophytes) or CPPD; MCP/wrist synovitis suggests RA. Isolated ankle or elbow OA is usually post-traumatic (secondary).

BOARD TRAP

WRONG answer: calling 2nd/3rd MCP arthropathy with hook-shaped osteophytes 'primary OA.' That pattern is the buzzword for HEMOCHROMATOSIS — order ferritin and transferrin saturation. The discriminator is the atypical-for-OA joint distribution (MCPs) prompting a secondary-cause workup.

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5. Heberden nodes (DIP)

WHAT YOU SEE

Hand with bony DIP nodules labeled HEBERDEN

WHAT IT ENCODES

Heberden nodes are hard, BONY enlargements (osteophytes) at the DIP joints; Bouchard nodes are the equivalent at the PIP joints. Mnemonic: alphabetical order matches anatomical order distal-to-proximal — B(ouchard) before H? No: Heberden = High/distal DIP; Bouchard = Below at PIP. They are firm and non-tender, reflecting bone, not the soft, warm, tender synovial swelling of RA.

BOARD TRAP

WRONG answer: interpreting firm DIP nodules as RA synovitis or as gouty tophi. Heberden/Bouchard nodes are BONY and represent osteophytes of OA. The discriminator: hard and bony = OA osteophytes; soft/boggy and tender = RA synovitis; chalky-white and may ulcerate = gouty tophi.

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6. Asymmetric / women after menopause

WHAT YOU SEE

'asymmetric', 'women >> after menopause', major risk

WHAT IT ENCODES

Hand OA is typically ASYMMETRIC and clusters in women, particularly after menopause (estrogen loss is implicated). Erosive (inflammatory) hand OA is a more aggressive subtype seen in postmenopausal women, producing a 'gull-wing' central erosion pattern on X-ray of the IP joints — but it still favors DIP/PIP, not MCP.

BOARD TRAP

WRONG answer: choosing RA because a postmenopausal woman has hand arthritis. OA is usually asymmetric and DIP-predominant; RA is symmetric and MCP/wrist/PIP-predominant. The discriminator: symmetric small-joint MCP/wrist synovitis with >1 hr stiffness and positive anti-CCP = RA; asymmetric DIP bony nodes with brief stiffness = OA.

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7. Obesity (knee) risk factor

WHAT YOU SEE

'3–6× deep knee', obesity/weight panel

WHAT IT ENCODES

Obesity is a leading MODIFIABLE risk factor, raising knee OA risk roughly 3–6 fold through increased mechanical load AND systemic adipokine-mediated inflammation (leptin, IL-6). Because the metabolic component matters, obesity also raises hand OA risk where load is minimal. Weight loss of even 5–10% measurably reduces knee pain and is a cornerstone of management.

BOARD TRAP

WRONG answer: dismissing weight loss as merely supportive. In overweight patients, weight reduction is a CORE, evidence-based intervention that lowers symptoms and can be curative in mild knee OA. The discriminator on a 'best initial management' question for an obese patient with knee OA is weight loss + exercise, not jumping to surgery or chronic opioids.

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8. Morning stiffness <30 min

WHAT YOU SEE

'STIFFNESS CLOCK', short-duration gel phenomenon

WHAT IT ENCODES

OA stiffness is brief — typically <30 minutes — and is a 'gel phenomenon' that eases quickly with movement. Pain is use-related: worse with activity and at the end of the day, better with rest. This contrasts with inflammatory arthritis, where stiffness lasts >1 hour, is worst in the morning, and improves with activity.

BOARD TRAP

WRONG answer: diagnosing OA in a patient with >60 minutes of morning stiffness and pain that improves with use. Prolonged morning stiffness (>1 hr) is the classic discriminator for INFLAMMATORY arthritis (RA, spondyloarthritis). The discriminator: short stiffness + activity-worsened pain = OA; prolonged stiffness + activity-relieved pain = inflammatory.

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9. Synovial fluid: non-inflammatory

WHAT YOU SEE

Fluid vial, 'SYNOVIAL FLUID', WBC low, non-inflammatory

WHAT IT ENCODES

OA synovial fluid is NON-inflammatory: clear/straw-colored, viscous (high hyaluronate), with WBC typically <2000/µL (and <25% PMNs). Inflammatory fluid (RA, gout) runs 2000–50,000/µL; septic fluid is usually >50,000/µL (often >100,000) with >90% PMNs. Always send fluid for cell count, crystals, and Gram stain/culture when the diagnosis is uncertain.

BOARD TRAP

WRONG answer: anchoring on OA and skipping arthrocentesis when a single joint is acutely hot and swollen. A WBC >50,000/µL or a positive Gram stain means SEPTIC arthritis — a surgical emergency, not OA. The discriminator is the synovial WBC: <2000 OA, 2000–50,000 inflammatory/crystal, >50,000 septic.

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10. X-ray: osteophytes, JSN, sclerosis, cysts

WHAT YOU SEE

X-ray panel: osteophytes, joint-space narrowing, subchondral sclerosis/cysts; 'correlates poorly with pain'

WHAT IT ENCODES

The four radiographic hallmarks of OA (mnemonic 'LOSS'): Loss of joint space (ASYMMETRIC narrowing, where cartilage is worn), Osteophytes (marginal bone spurs), Subchondral Sclerosis, and Subchondral cysts (geodes). Notably ABSENT are the periarticular osteopenia and marginal erosions of RA. Plain weight-bearing films are the standard initial imaging — MRI is rarely needed.

BOARD TRAP

WRONG answer: escalating treatment based on severe X-ray changes in a minimally symptomatic patient. Radiographic severity correlates POORLY with pain — treat the patient, not the film. The discriminator: ASYMMETRIC joint-space narrowing + osteophytes + sclerosis = OA; SYMMETRIC narrowing + periarticular osteopenia + marginal erosions = RA.

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11. Knee-shaped rusty car / spurs

WHAT YOU SEE

Rusty 'KNEE-SHAPED CAR' on lift, SPURS labels

WHAT IT ENCODES

The rusting car is the joint with progressive cartilage loss; the spurs welded onto it are osteophytes — reactive new bone at joint margins. This captures the dual nature of OA: cartilage is destroyed AND bone proliferates (osteophytes, sclerosis). Contrast with RA, where bone is eroded/destroyed (marginal erosions) rather than built up.

BOARD TRAP

WRONG answer: treating OA as a primary autoimmune synovitis and starting DMARDs/biologics. OA has no role for methotrexate or TNF inhibitors; management is weight loss, exercise, topical/oral NSAIDs, then intra-articular steroids, then joint replacement. The discriminator: bone-building (osteophytes) degenerative disease = OA, not erosive autoimmune disease = RA.

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12. Pathology: cartilage loss, MMPs

WHAT YOU SEE

Cartilage diagram, MMPs, chondrocyte changes, 'PATHOLOGY'

WHAT IT ENCODES

OA is an ACTIVE metabolic process, not passive wear: chondrocytes shift to a catabolic phenotype and release matrix metalloproteinases (MMP-13/collagenase, aggrecanases/ADAMTS) plus IL-1 and TNF, degrading type II collagen and proteoglycans. The cartilage loses its elastic recoil; the joint fibrillates and ulcerates; subchondral bone remodels with sclerosis and cysts. Early there is paradoxical cartilage swelling before net loss.

BOARD TRAP

WRONG answer: labeling OA 'just wear and tear' with no biology. The MMP/cytokine-driven cartilage degradation is the reason OA is increasingly viewed as a disease of the whole joint. The discriminator for pathology questions: OA = MMP/aggrecanase cartilage breakdown with subchondral remodeling; RA = pannus (proliferative inflamed synovium) eroding cartilage and bone.

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13. Charcot joint (exception)

WHAT YOU SEE

'CHARCOT JOINT', rapid severe destruction, neuropathy

WHAT IT ENCODES

Charcot (neuropathic) arthropathy is a destructive arthropathy from LOSS OF PROTECTIVE SENSATION: repeated unperceived microtrauma causes rapid, severe joint destruction, fragmentation, and deformity. Most common cause today is DIABETIC neuropathy affecting the foot/midfoot (rocker-bottom deformity); historically tabes dorsalis (neurosyphilis) and syringomyelia. The classic clue is destruction grossly out of proportion to the patient's (often minimal) pain.

BOARD TRAP

WRONG answer: calling a grossly destroyed, deformed, but relatively painless joint in a diabetic 'advanced OA' or 'septic arthritis.' Painless rapid destruction + peripheral neuropathy = CHARCOT joint. The discriminator is the dissociation between severe radiographic destruction and minimal pain, in a patient with neuropathy.